Al Imam Mohammad Ibn Saud Islamic University
College of Computer and Information Sciences

Information Systems Department

(Principals of Project Management)

Course Project

Course Title:	Principals of Project Management
Course Code:	IS 1241
Semester/Year:	1st Sem 2025-26
Course Instructor:	Dr. Majed A Albraithen
Due Date	Sunday 30th of November

Group [3]

Section [171]

Student #	Student Name	Student ID	Responsibilities	Mark 20
(Leader) Student 1	***Abdullah Nasser ALHussain***	444003463	**Task2**	
Student 2	***khalid waleed alzamil***	445011804	**Task3**	
Student 3	***Mohammed ahmed albazaee***	444010694	**Task1**	
Student 4	***Abdullah Sami Alahmed***	443014578	**Task2**	
Student 5	***Abdullah Omar Alyabis***	443016043	**Task 1**	
Student 6	***Yasser Ibrahim alzughabi***	444001862	**Task 3**	

Graded Item	5	4.5	4	3.5	3	2.5
WBS 1. Well explained approach 2. explained the bases of major work items 3. correct overall WBS structure 4. included development and PM processes 5. correct WBS item description	All items clear and correct.	**One** item **not well** done.	**two** items **not well** done. **Or** *one item not provided*.	**three** items **not well** done. Or *one item not provided* and **one not well** done.	**Four items not well done.** **Or** *two items not provided*. Or **two items not well** done and *one item not provided*.	**Five items not well done.** **Or** *two items not provided* and **one not well** done. Or *one not provided* and **three not well** done.
Activity list and attributes Well-developed activity list and attributes showing: 1. activity ID and name 2. well estimated activity duration 3. well determined **dependencies, predecessors, constraints, and assumptions** 4. Well-developed Milestone list 5. consistent with WBS and project charter	All items clear and correct.	**One** item **not well** done.	**two** items **not well** done. **Or** *one item not provided*.	**three** items **not well** done. Or *one item not provided* and **one not well** done.	**Four items not well done.** **Or** *two items not provided*. Or **two items not well** done and *one item not provided*.	**Five items not well done.** **Or** *two items not provided* and **one not well** done. Or *one not provided* and **three not well** done.
Project schedule using MS project Well developed project schedule showing: 1. consistency with activity list and attributes. 2. schedule view with outline numbers. 3. milestones on schedule. 4. milestones on Gantt Chart. 5. And correctly defined (milestones, dependencies, durations) ***Please clearly referenced chosen tool**	All items clear and correct.	**One** item **not well** done.	**two** items **not well** done. **Or** *one item not provided*.	**three** items **not well** done. Or *one item not provided* and **one not well** done.	**Four items not well done.** **Or** *two items not provided*. Or **two items not well** done and *one item not provided*.	**Five items not well done.** **Or** *two items not provided* and **one not well** done. Or *one not provided* and **three not well** done.
Network Diagram 1. NW diagram view 2. Showing activities 3. Showing dependencies 4. Showing critical path 5. Fit in one figure	All items clear and correct.	**One** item **not well** done.	**two** items **not well** done. **Or** *one item not provided*.	**three** items **not well** done. Or *one item not provided* and **one not well** done.	**Four items not well done.** **Or** *two items not provided*. Or **two items not well** done and *one item not provided*.	**Five items not well done.** **Or** *two items not provided* and **one not well** done. Or *one not provided* and **three not well** done.

IS1241 - Project Evaluation Form (1st Sem 2025-26)

Group number			
Criteria		Mark	Obtained
Task 1:		**5**	0
1.	WBS	5	0
2.	Activity list and attributes and milestone list	5	0
Task 2:		**5**	0
3.	Project schedule using MS project	5	0
4.	Network Diagram using MS project	5	0
Task 3:		**5**	0
5.	Presentation Design (content presentation effectiveness)	5	0
6.	Organized presentation logically (coherence)	5	0
7.	well written report	5	0
Discussion		**5**	0
1.		5	0
2.		5	0
3.		5	0
4.		5	0
5.		5	0
6.		5	0
Total mark			
Team members:			Mark out of 20
1.			0
2.			0
3.			0
4.			0
5.			0
6.			0

1. Creating WBS

In order to create the WBS on the Al-Bawadi Digital Transformation Initiative, two methodologies of the slides were used:

(1) Deliverable-Based Approach

Scope definition and WBS which states that WBS should break down deliverables into manageable elements.
Our deliverables include:

-Digital Asset Management System.
-AR Mobile Application
-Membership System Ticketing
-Admin Dashboard
-Database & Cloud Infrastructure.
-Training & Documentation

Project management process group approach to project management is a method that has been developed and used by numerous large and small firms to handle projects. Project Management Process Group Approach Project management process group approach to project management is a strategy that has been developed and applied by several large and small firms in managing projects.
We matched the key WBS levels and the PMBOK phases:

-Initiation
-Planning
-Execution (Development)
-Monitoring & Controlling
-Closing
-Foundations of Classification of Work.

The classification of the WBS was done according to:

-Product deliverables
-Processes of project management.
- Project life cycle phases

This makes it complete and totally traceable, in addition to being in full compliance with the Project Charter.

1.2 WBS Tabular Form

WBS ID	Work Package / Task
1.0	Project Initiation
1.1	Approve Project Charter
1.2	Assign Project Roles & Responsibilities
1.3	Stakeholder Identification
2.0	Planning Phase
2.1	Develop Project Management Plan
2.2	Develop Scope Statement
2.3	Communication & Risk Plans
2.4	Requirements Gathering Sessions
3.0	System Analysis & Design
3.1	Requirements Analysis
3.2	System Architecture Design
3.3	Database Design (Cloud)
3.4	UI/UX Design (AR App, Ticketing App, Dashboard)
4.0	System Development
4.1	Digital Asset Management System
4.1.1	Asset upload & metadata module
4.1.2	Cloud database integration
4.2	Mobile Application with AR
4.2.1	Mobile App Development
4.2.2	AR Feature Integration
4.3	Ticketing & Membership Web System
4.4	Admin Dashboard Development
5.0	Testing
5.1	Unit Testing
5.2	Integration Testing
5.3	User Acceptance Testing (UAT)
6.0	Deployment & Training
6.1	Staff Training Workshops
6.2	User Manual Preparation
6.3	System Deployment
7.0	Closing Phase
7.1	Final Handover Report
7.2	Lessons Learned
7.3	Final Sign-Off

1.3 WBS Dictionary for One Work Package Item

Item	Description
Work Package Name	AR Feature Integration
Purpose	Integrate Augmented Reality (AR) capabilities into the museum mobile app to allow users to scan artifacts and view 3D models, animations, and enhanced digital content.
Scope	Develop AR scanning module, integrate with backend, ensure cross-platform support (iOS/Android).
Inputs	3D Models, UI/UX AR mockups, AR SDK documentation.
Deliverables	Fully functional AR interaction module within the mobile application.
Success Criteria	Smooth scanning, high accuracy of object detection, AR overlay rendered without delay, passes UAT.
Assumptions	3D assets available and approved by museum curators.
Constraints	Must follow museum content security requirements.

2. Define Activities, Sequence Activities and Estimate Activity Durations

This section presents the detailed activity list and attributes for the *Al-Bawadi Digital Transformation Initiative*. All activities are derived from the approved WBS and project charter and are assigned unique IDs and names to ensure clear traceability. For each activity, realistic durations have been estimated using expert judgment and the typical effort required for similar software development tasks. Dependencies, predecessors, constraints, and assumptions are explicitly defined to reflect the logical execution order of the project and to provide a solid basis for scheduling, risk analysis, and subsequent development of the Gantt chart and network diagram.

2.1 Activity List and Attributes

ID	Activity Name	Duration (Days)	Predecessor (s)	Dependency Type	Constraints	Assumptions
1	Approve Project Charter	3	—	—	Must be completed before any planning work starts	Sponsor and management are available to review and approve the charter
2	Assign Project Roles	2	1	FS: 1 → 2	Roles must be aligned with available staff and budget	Required team members are available and willing to participate
3	Identify Stakeholders	3	1	FS: 1 → 3	Stakeholder list must be completed before requirements sessions	Stakeholders can be identified using existing museum records and contacts
4	Requirements Gathering	8	2, 3	FS: 2/3 → 4	Workshops must be scheduled within museum working hours	Stakeholders will attend sessions and provide necessary information
5	Develop Scope Statement	4	4	FS: 4 → 5	Scope must be approved before detailed planning and design	Requirements are stable enough to define project boundaries
6	Develop PM, Communication & Risk Plans	6	5	FS: 5 → 6	Plans must follow university and museum policies	Project manager has access to needed templates and guidelines
7	Requirements Analysis	6	4	FS: 4 → 7	Analysis must be completed before design can start	Requirements gathered are complete and documented
8	System Architecture Design	10	7	FS: 7 → 8	Must comply with technical standards and security policies	Selected technologies (cloud, AR, web) are feasible for the museum
9	UI/UX Design	12	7	FS: 7 → 9	Must follow usability and accessibility guidelines	Users will provide feedback on early designs if needed
10	Database Schema Design	7	8	FS: 8 → 10	Must be compatible with chosen cloud platform	Data formats and entities are clearly defined during analysis
11	Asset Management System Development	15	10	FS: 10 → 11	Development must use approved tools and standards	Development team is familiar with web/backend technologies

ID	Activity Name	Duration (Days)	Predecessor (s)	Dependency Type	Constraints	Assumptions
12	Mobile App Development	20	9	FS: 9 → 12	Must support required mobile platforms (Android/iOS)	Mobile development tools and licenses are available
13	AR Module Integration	12	12	FS: 12 → 13	AR library/SDK must be compatible with the app	AR content (3D models, images) is provided and approved
14	Ticketing System Development	14	10	FS: 10 → 14	Must integrate with payment or ticketing policies of the museum	Functional requirements for ticketing are clear and stable
15	Admin Dashboard Development	10	10	FS: 10 → 15	Must provide required reports and control functions	Admin roles and permissions are defined during analysis
16	Unit Testing	8	11, 12, 13, 14, 15	FS: 11/12/13/14/15 → 16	All modules must reach a testable state before testing starts	Test environment and test data are available
17	Integration Testing	6	16	FS: 16 → 17	All key interfaces between systems must be implemented	No major architectural change during integration
18	User Acceptance Testing	5	17	FS: 17 → 18	UAT must be done before deployment	Museum staff and management are available for UAT sessions
19	Staff Training	4	18	FS: 18 → 19	Training must be scheduled without disrupting museum operations	Staff are willing to adopt the new digital systems
20	User Manual Preparation	3	18	FS: 18 → 20	Manuals must reflect the final accepted system	UAT feedback is available to refine documentation
21	System Deployment	5	18	FS: 18 → 21	Deployment should be done in a low-traffic period	Infrastructure and hosting environment are ready
22	Handover Report	5	21	FS: 21 → 22	Report must summarize all deliverables and status	Project documentation is complete and accessible
23	Lessons Learned	3	21	FS: 21 → 23	Must be completed before final sign-off	Team members are available to share feedback and insights
24	Final Sign-Off	1	22, 23	FS: 22/23 → 24	Sponsor must review handover and lessons learned	Sponsor is satisfied with project outcomes and deliverables

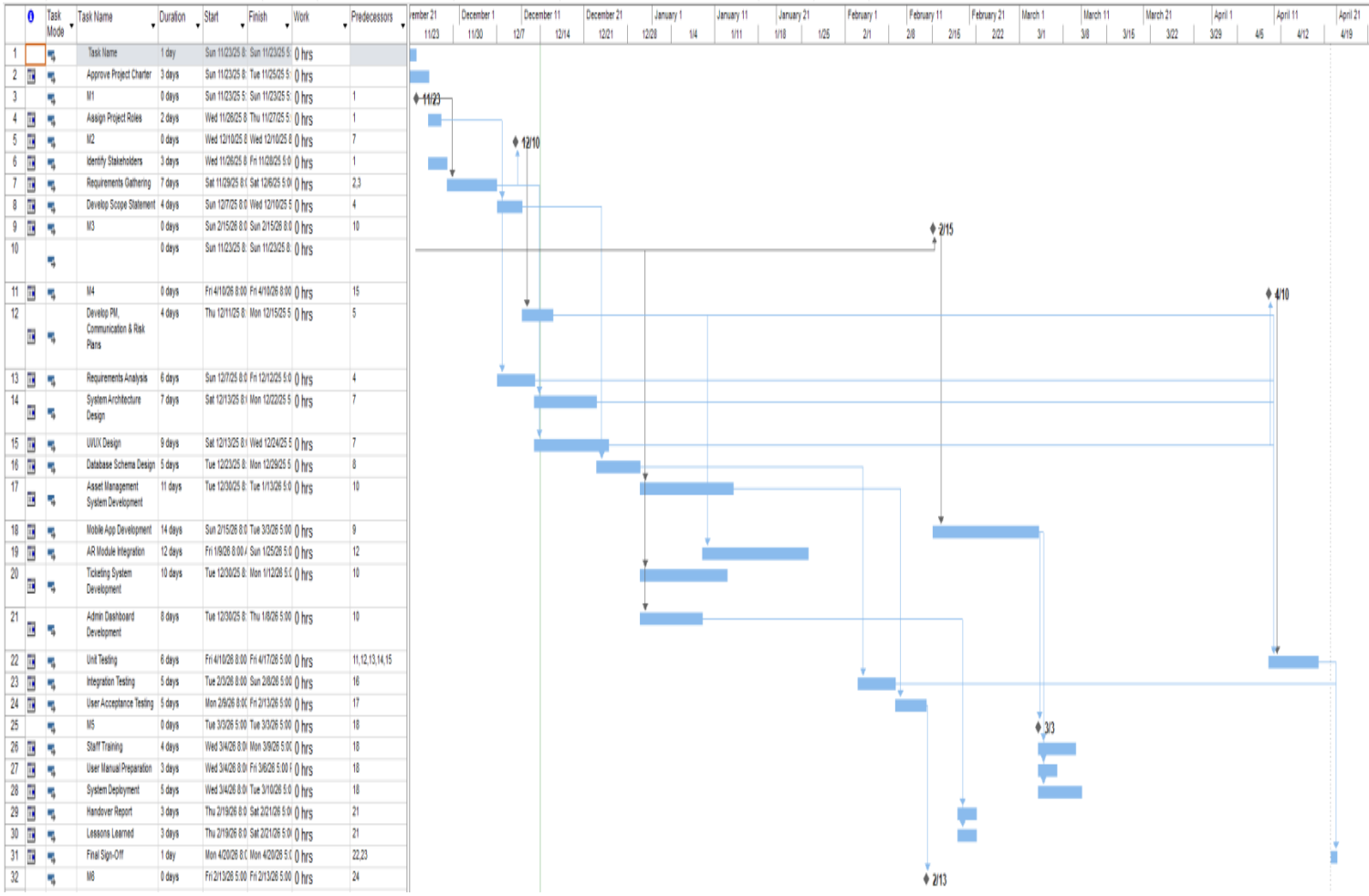
2.2 Milestone List

Milestones represent **major project achievements** and have **zero duration**

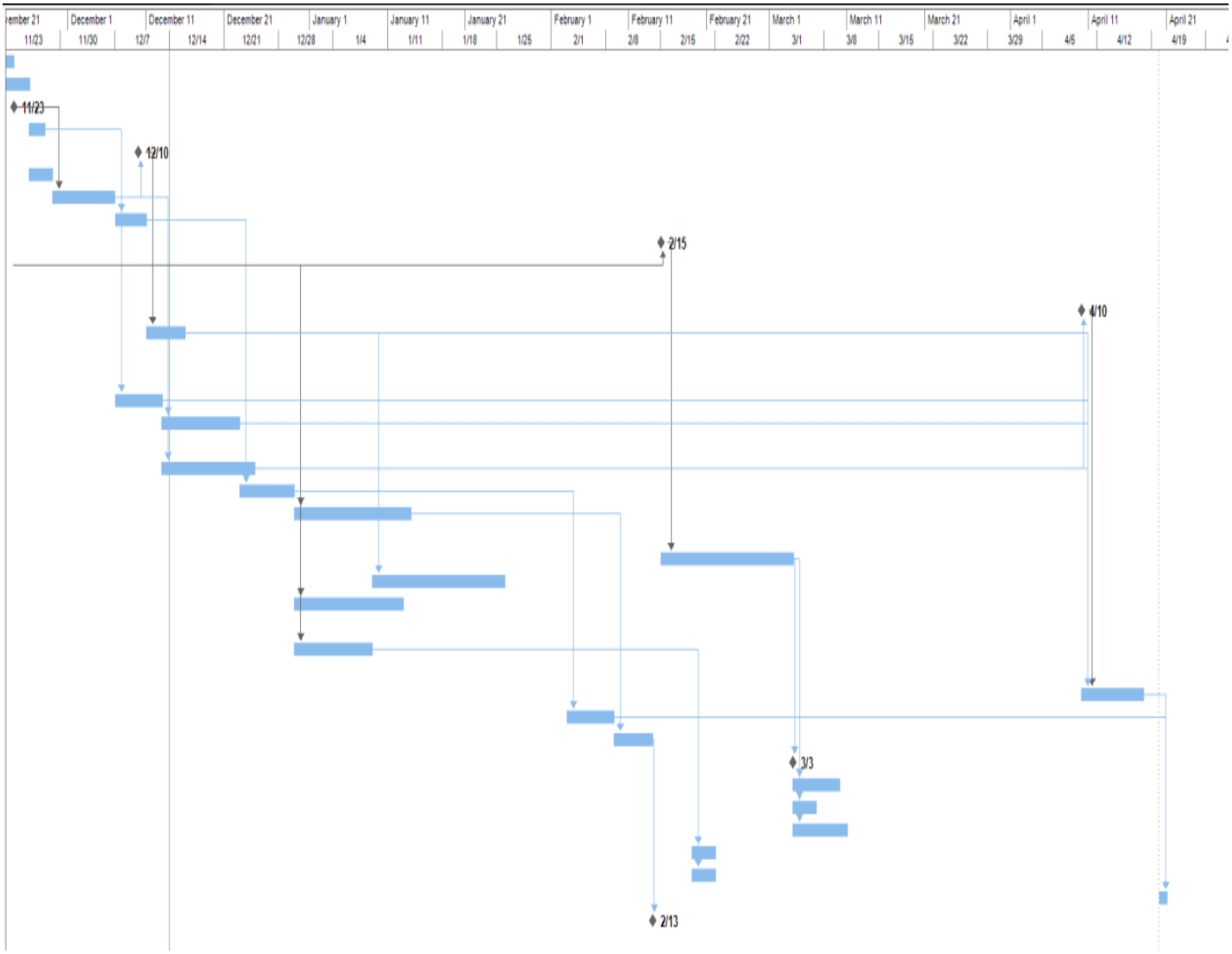
Milestone ID	Milestone Name	Description	Date
M1	Project Charter Approved	Authorization to begin project	23 Nov 2025
M2	Requirements Completed	Completion of requirement gathering & analysis	10 Dec 2025
M3	System Architecture & Design Completed	Finalization of UI/UX + architecture + DB	15 Feb 2026
M4	Prototype Development Completed	Initial working version ready for evaluation	10 Apr 2026
M5	Quality Assessment Completed	Completion of major testing efforts	20 Jun 2026
M6	Final Implementation & Closure	Full deployment + handover complete	23 Jul 2026

3. Develop Project Schedule Using MS Project (MS Project)

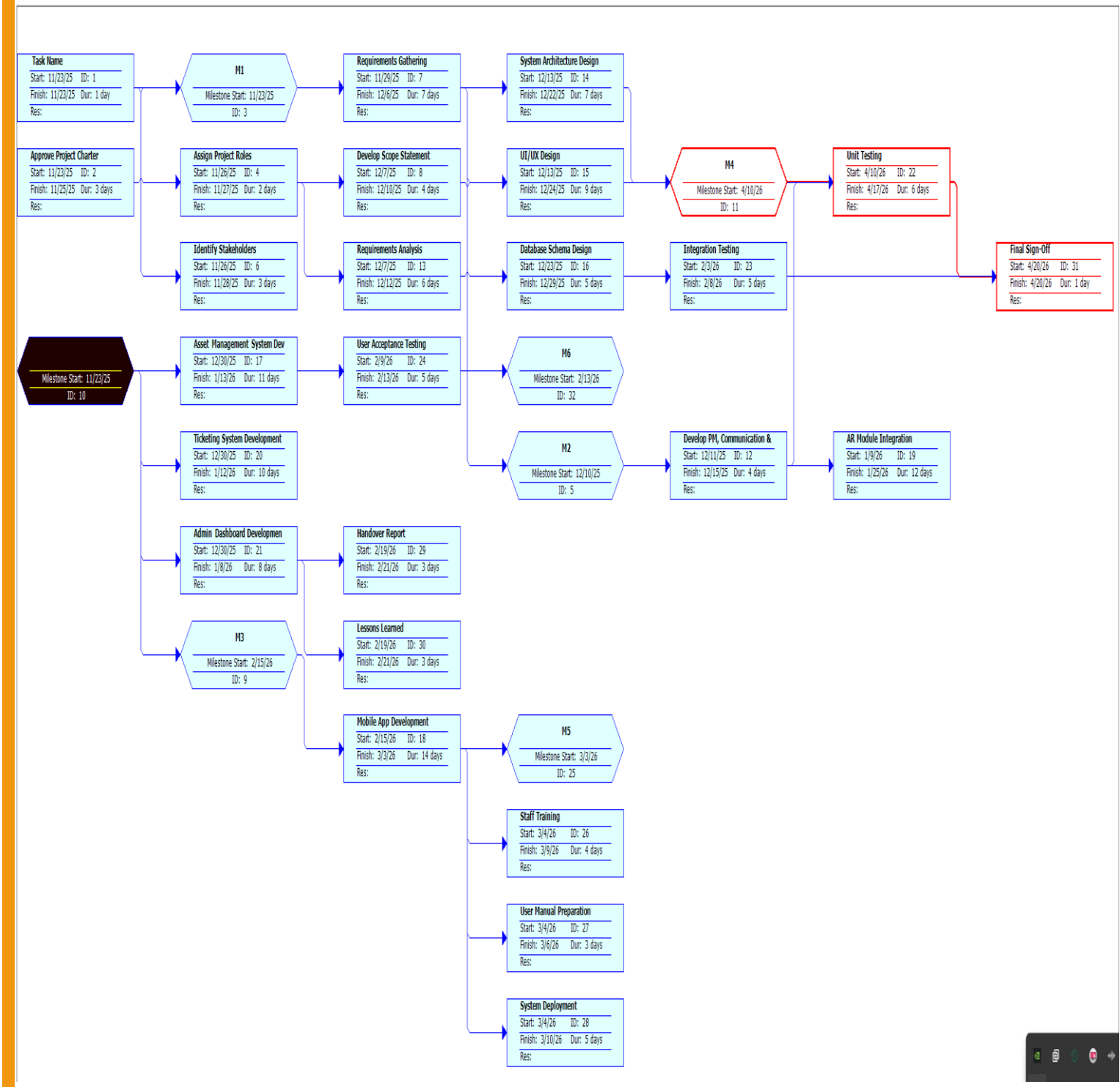
3.1 Milestones on Schedule View



3.2 Milestones on Gantt Chart



3.3 Network Diagram



Reference:

IS1241 Course Materials, “Principles of Project Management “

Microsoft Project viewer 365

Appendix:

Peer review form

For each team member, rate their performance on a scale of 1 to 5 for each criterion.

- **5 - Exceptional: Consistently goes above and beyond, exemplifying this behavior.**
- **4 - Good: Consistently meets expectations and contributes positively.**
- **3 - Satisfactory: Meets minimum expectations but could improve.**
- **2 - Developing: Inconsistent and sometimes fails to meet expectations.**
- **1 - Needs significant improvement: Regularly fails to meet expectations and negatively impacts the team.**

	S1	**S2**	**S3**	**S4**	**S5**	**S6**
Communication	**5**	**5**	**5**	**5**	**5**	**5**
Effectively communicates ideas to the team	**5**	**5**	**5**	**5**	**5**	**5**
Actively listens to and values others' input	**5**	**5**	**5**	**5**	**5**	**5**
Communicates professionally and respectfully	**5**	**5**	**5**	**5**	**5**	**5**
Consistently meets deadlines and commitments	**5**	**5**	**5**	**5**	**5**	**5**
Contributes a fair and equal share of the workload	**5**	**5**	**5**	**5**	**5**	**5**
Handles disagreements constructively and professionally	**5**	**5**	**5**	**5**	**5**	**5**